Jaidhar Ramagiri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.79
Naarayana Junior College	Class XII (TSBIE)	2023	9.33%
Seventh Day Adventist High School	Class X (SSC)	2021	9.80%

INTERNSHIP EXPERIENCE

Research Intern - IIT Kharagpur

Inverse Problems in Pattern Formation Using the ODIL Framework | Supervisor: Dr. Saikat Mukherjee | [GitHub]

- Reproduced four forward and inverse PDE benchmarks using **ODIL (Optimizing a Discrete Loss)**, validating discrete PDE residual optimization and demonstrating **fewer optimization iterations** than **PINN-based approaches**, with faster convergence using Newton-based optimization.
- Generated synthetic Cahn-Hilliard phase-separation data using **FiPy** and recovered the **Flory Huggins interaction parameter** (= 2.1-3.0) through ODIL-based inverse modeling with L-BFGS-B, Adam, and Newton optimization.
- Developing a hybrid **ODIL-Neural Network** framework to learn the unknown non-ideal free energy contribution **directly** from concentration **evolution data**, replacing the analytical relation and enabling **data-driven free energy discovery**.

PROJECTS

Solving Ideal MHD Induction Equation Using PINNs | B.Tech Honors Project II | [GitHub]

- Developed a **PyTorch-based Physics-Informed Neural Network (PINN)** using automatic differentiation and physics-based loss functions to solve coupled PDEs.
- Experimented with **neural network architectures, Fourier feature embeddings, and loss formulations**, analyzing their impact on model performance and generalization.
- Evaluated predictions against **analytical and RK4-spectral numerical solutions** using L2 error analysis and training-loss curves to investigate accuracy and error evolution.

Physics-Informed Neural Networks for PDEs | B.Tech Project I | [GitHub]

- Implemented PyTorch **PINNs** for Burgers, Riemann, 3D heat, and exploratory dynamo models.
- Studied effects of **activations, loss weighting**, and **architecture** on solution stability.
- **Identified limitations** of classical PINNs for vector PDEs and built a compact **training and visualization pipeline**.

Bayesian Cosmological Inference & GRB LIV Analysis | BTech Project II | [GitHub]

- Analyzed **32 cosmic chronometer H(z) measurements** using **Bayesian likelihood modeling** and **MCMC-based parameter estimation** under flat and open CDM cosmological models.
- Reproduced a **hierarchical Bayesian inference framework** for **32 Gamma-Ray Bursts (GRBs)**, implementing intrinsic spectral-lag models and population-level parameter estimation using **dynesty** and **bilby**.
- Performed **posterior analysis** and **uncertainty quantification** by comparing intrinsic lag models and population distributions to investigate constraints on **Lorentz Invariance Violation (LIV)**.

Stars, Galaxies, and Quasars Classification Model | Python | [GitHub]

- Achieved 99% classification accuracy by **building** multiple ML models (**Decision Tree**, **Logistic Regression**, **KNN**, **SVM**) using **Sloan Digital Sky Survey** astronomical data.
- Conducted data preprocessing, feature selection, and model evaluation using Python, Pandas, NumPy, and Scikit-learn.

POSITIONS OF RESPONSIBILITY

Head - PR & Outreach, Prayas | Social Service Club | IIT Hyderabad

- Led outreach and publicity for community-education initiatives; coordinated school engagement and promotional campaigns for **Prerna 2025**, which hosted **500+ rural students, parents, and teachers** at IITH.

Mentor, Sunshine Club | Mental Health Council | IIT Hyderabad

- Mentored **15+ freshmen** and supported student engagement and awareness activities.

Core Member, Cepheid | Astronomy Club | IIT Hyderabad

- Organized astronomy awareness sessions and outreach activities.

Core Member, Aero Club | IIT Hyderabad

- Participated in aeromodelling, RC aircraft, glider, and aerodynamic activities.

RELEVANT COURSES

Mathematics: Calculus I & II, Elementary Linear Algebra, Differential Equations, Introduction to Probability, Introduction to Statistics, Advanced Mathematical Physics, Probability and Random Variables

CSE & DS: Introduction to Programming, Data Structures & Applications, Artificial Intelligence, Data Science Analysis, Machine Learning for Physical Sciences, *Machine Learning for High Energy Physics

SKILLS

Programming Languages: C, C++, Python, MATLAB, SQL* and JavaScript*

Libraries & Tools: Jupyter, NumPy, Matplotlib, Pandas, Scikit-learn, PyTorch, Seaborn, emcee, dynesty, bilby.

Soft Skills: Leadership, Team Management, Effective Communication, Problem Solving

EXTRACURRICULAR

NSS Actively participated in the social events conducted by the NSS community in IITH

Engaged in recreational sports, chess, and drumming; also passionate about manhwas and storytelling